

Aggressive B-cell lymphoma

DLBCL and the three situations that change the plan — **Burkitt, primary CNS lymphoma**, and **HIV-associated** disease. The recurring questions are cell of origin, double-hit status, cardiac fitness, and whether the CNS needs covering

Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin, pp.194–222 · items marked † are outside the handbook chapter

WORKUP

- **Excisional lymph node biopsy preferred**, core acceptable, **no FNA**; hematopathology review
- IHC: CD20, CD10, CD30, CD5, BCL2, BCL6, MUM1, MYC, EBV + **Ki-67**. Cell of origin by **Hans algorithm** (GCB vs non-GCB/ABC)
- **Cytogenetics for MYC, BCL2, BCL6** — defines double/triple hit (HGBCL)
- Labs: CBC, CMP, **LDH**, uric acid, **hepatitis B and C serologies, HIV**
- **PET/CT + bone marrow biopsy**; baseline **echo or MUGA**
- Situational: LP if CNS concern · endoscopy if GI concern · **fertility preservation**
- Response: PET/CT at completion, marrow biopsy if positive at diagnosis

IPI — AGE >60 · PS ≥2 · LDH ↑ · ≥2 EXTRANODAL · STAGE III–IV

Risk	5-yr OS pre-R	3-yr OS post-R
Low (0–1)	73%	91.4%
Low-intermediate (2)	51%	80.9%
High-intermediate (3)	43%	65.1%
High (4–5)	26%	59%

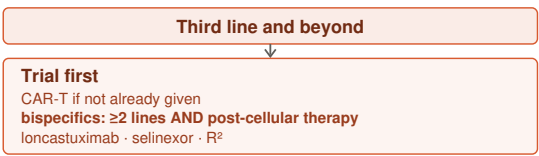
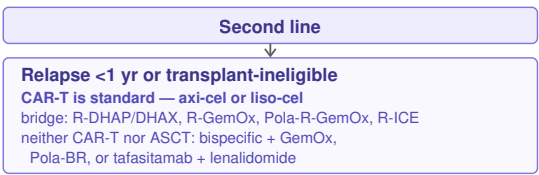
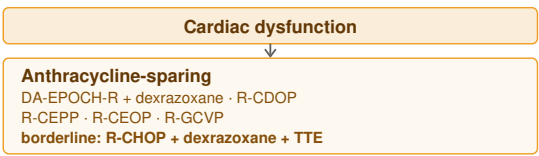
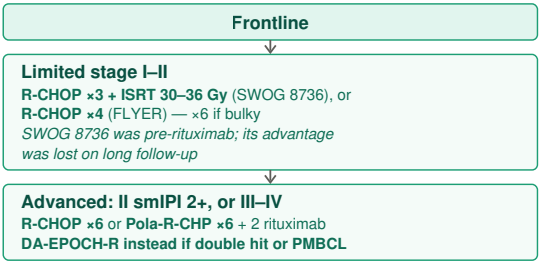
Mnemonic **APLES**, one point each. Shipp, NEJM 1993 · Ziepert, JCO 2010. The handbook gives neither **R-IPI** nor **NCCN-IPI**.

CNS RISK AND PROPHYLAXIS

CNS-IPI	IPI + kidney and/or adrenal (0–6). 2-yr CNS relapse: 0–1 = 0.6% · 2–3 = 3.4% · 4–6 = 10.2%
Who gets it	Mandatory: Burkitt/ALL, testicular. High risk: CNS-IPI 4–6 · double/triple hit and non-GCB · epidural, paranasal sinus or base of skull · breast or uterine
Regimen	Systemic HD-MTX 3–3.5 g/m² x2–4 after completing systemic therapy — preferred in fit patients with adequate renal function for parenchymal penetration. Or IT MTX ± cytarabine x4–8

The handbook calls the benefit **controversial** (Wilson, Blood 2022) — consider in selected high-risk patients, not routinely.

TREATMENT PATHWAY



Recurring supportive care. Antiviral + PJP prophylaxis with nearly every regimen: **acyclovir 400 mg BID + Bactrim DS three times weekly** (or SS daily). **G-CSF** with R-CHOP if age ≥60, required with Pola-R-CHP. **Cytarabine → prednisolone 1% eye drops** 2 gtt OU BID from 12 h before until 48 h after; reduce cytarabine to 1 g/m² if age >60. **Mesna with every ifosfamide.**

FRONTLINE REGIMENS

Regimen	Dosing	Cycle
R-CHOP	Rituximab 375 · cyclophosphamide 750 · doxorubicin 50 · vincristine 1.4 (max 2 mg) all d1; prednisone 40/day PO d1–5	q21d x6
Pola-R-CHP	Polatuzumab vedotin 1.8 mg/kg · rituximab 375 · cyclophosphamide 750 · doxorubicin 50 all d1; prednisone 100 mg PO d1–5	q21d x6, +2 rituximab
DA-EPOCH-R	Rituximab 375 d1; doxorubicin 10/day + etoposide 50/day + vincristine 0.4/day (max 2 mg) all CIV d1–4; cyclophosphamide 750 d5; prednisone 60/day PO d1–5	q21d x6
R-CEOP	Cyclophosphamide 750 + vincristine 1.4 d1; etoposide 50 IV d1 then 100 PO d2–3; prednisone 40/day d1–5	q21d x4–6
R-CEPP	Cyclophosphamide 600 d1; etoposide 70 d1–3; prednisone 60 + procarbazine 60 PO d1–10	q28d x6
R-GCVP	Gemcitabine 1000 d1+8; cyclophosphamide 750 + vincristine 1–1.4 d1; prednisolone 100 mg PO d1–5	q21d

All mg/m² unless stated. **DA-EPOCH-R escalation** (etoposide, doxorubicin, cyclophosphamide; CBC twice weekly): ANC nadir ≥500 throughout → +20%; <500 on 1–2 checks → same; <500 on 3 checks or platelets <25k → –20%.

SALVAGE AND LATER-LINE

Regimen	Dosing	Cycle
R-ICE	Ifosfamide 5000 + mesna 5000 CIV d2; carboplatin AUC 5 (max 800 mg) d2; etoposide 100 d1–3	q14–21d
R-DHAP · R-DHAX	Cytarabine 2000 q12h x2 d2 + dexamethasone 40 mg d1–4, with cisplatin 100 (DHAP) or oxaliplatin 100 (DHAX) d1	q21–28d
R-ESHAP	Methylprednisolone 500 mg d1–5; etoposide 40 d1–4; cisplatin 25/day CIV d1–4; cytarabine 2000 d5	q21–28d
R-GDP · R-GCD	Gemcitabine 1000 d1+8; dexamethasone 40 mg PO d1–4; cisplatin 75 d1 (GDP) or carboplatin AUC 5 (GCD)	q21d
R-GemOx	Gemcitabine 1000 + oxaliplatin 100 d1	q14–21d x6
Pola-BR	Polatuzumab 1.8 mg/kg + rituximab 375 d1; bendamustine 90 d1+2	q21d x6
Tafasitamab + len	Tafasitamab 12 mg/kg d1; lenalidomide 25 mg PO d1–21	q28d x12
Epcoritamab	SQ step-up: 0.16 mg C1D1 → 0.8 mg C1D8 → 48 mg C1D15 , then weekly to C3, q2wk C4–9, q4wk C10+	q28d
Glofitamab	Obinutuzumab 1000 mg C1D1 pretreatment, then 2.5 mg C1D8 → 10 mg C1D15 → 30 mg C2D1	q21d x12

Transplant candidates get 2–3 cycles of platinum salvage; non-transplant 4–6. **Cytarabine → 1 g/m² if age >60**. Either bispecific may be given with GemOx.

PEARLS

- **DA-EPOCH-R, not R-CHOP, for double-hit HGBCL and primary mediastinal B-cell lymphoma**
- **R-CEOP outcomes appear inferior to R-CHOP** (Puckrin 2022) — for borderline cardiomyopathy prefer R-CHOP + dexrazoxane with echo surveillance over switching regimen
- **CAR-T now precedes transplant** in second line for early relapse — platinum salvage is bridging, not destination therapy
- **No MTX with effusions or ascites**, or with PPIs, penicillins, TMP-SMX, salicylates, probenecid or NSAIDs

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STAGING — COTSWOLDS-MODIFIED ANN ARBOR

I	One lymph node region or lymphoid structure (spleen, thymus, Waldeyer ring)
II	≥2 regions, same side of the diaphragm. Hilar nodes are lateralized — bilateral involvement is stage II
III	Regions on both sides of the diaphragm
IV	Disseminated — marrow, CNS, or more than one extranodal site
A / B	B = weight loss >10% over 6 mo, or unexplained fever >38°C in the past month, or recurrent drenching night sweats
S · X · E	S splenic · X bulky: node ≥10 cm , or mediastinal mass ≥½ the transverse thoracic diameter at T5–6 · E limited direct extranodal extension

A single extranodal site as the only disease is IE; more than one extranodal site is stage IV. Hilar (bronchopulmonary) nodes sit outside the mediastinum.

BURKITT — RISK AND PARAMETERS

Low risk	ALL of: normal LDH, ECOG 0–1, stage I–II, no mass ≥10 cm, ≤1 extranodal site
High risk	Everything else
Baseline	HIV, HBsAg, HBCab, HBsAb, HCV Ab · echocardiogram for EF
To start a cycle	ANC >1000, platelets >75,000, direct bili <1.4, Cr <1.2 — does not apply to cycle 1, nor to the day-10 HD-MTX, which is given regardless of counts

Also staged by St. Jude/Murphy. Every Burkitt patient gets CNS prophylaxis.

BURKITT PATHWAY

